

Hemant

Aspiring Software Development Engineer and AI/ML enthusiast with experience building scalable web applications and computer vision systems. Skilled in full-stack development, machine learning, and real-time object detection using modern AI frameworks.

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Portfolio: portfolio-hemant.vercel.app | GitHub: github.com/hemantmeena2005

Education

ABV-Indian Institute of Information Technology and Management <i>Integrated B.Tech and M.Tech in Information Technology</i>	CGPA: 7.44 2022 – 2027
Nalanda Modern Public School <i>CBSE Class XII</i>	Percentage: 90% 2022

Experience

AI/ML Intern – Defence Research and Development Organisation (DRDO), New Delhi – Worked on aerial vehicle and drone detection systems using YOLO -based object detection models. – Developed computer vision pipelines for real-time detection and tracking using OpenCV and deep learning techniques. – Performed model training, preprocessing, and inference optimization for improved detection accuracy on video datasets. – Collaborated on AI/ML research tasks focused on defense-oriented surveillance applications.	May 2026
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Skills

Languages: Python · JavaScript (ES6+) · Java · C++ · TypeScript
AI/ML & Computer Vision: Machine Learning · Deep Learning · OpenCV · YOLO · PyTorch · Computer Vision
Web Development: Node.js · React.js · Express.js · Next.js · Tailwind CSS · HTML5/CSS3
Databases: MongoDB · Appwrite · MySQL · PostgreSQL
Cloud & DevOps: AWS · Docker · Vercel
Tools: Git · GitHub · GSAP · Rapid API · Clerk Auth

Projects

Invoicely Next.js 14 , TypeScript , MongoDB , Stripe – Architected a type-safe SaaS platform using Next.js 14 for automated client billing and financial management. – Integrated Stripe API and email automation to reduce manual invoicing overhead by 60% . – Deployed secure OAuth authentication and real-time analytics dashboards to monitor platform performance.	June 2025
AI Recipe Generator Next.js , Tailwind CSS , Cohere API – Built an interactive assistant utilizing the Cohere API for real-time culinary recommendations. – Built a mobile-first, responsive interface with Tailwind CSS focused on high-speed user interactions. – Optimized system resilience via asynchronous error-handling, reducing failed API requests by 35% .	May 2025
ChatMate Next.js , Appwrite , Tailwind CSS – Engineered a full-stack social media application featuring instant messaging and real-time content sharing. – Leveraged Appwrite for serverless backend services, managing user authentication and media storage. – Improved database query performance and data fetching logic to improve server response times by 30% .	July 2024

Summer Projects

Explainable AI for Melanoma Detection Python , PyTorch , Computer Vision – Created a CAD system using a Convolutional Autoencoder to extract features for melanoma classification. – Engineered an Attention-based Network to prioritize critical lesion characteristics, achieving 90% accuracy . – Enhanced transparency by implementing visualization layers, addressing the black-box challenge in medical AI.	Summer 2025
Automatic Traffic Violation Detection OpenCV , YOLO , Python – Designed a computer vision system for real-time traffic monitoring and identification of vehicle violations. – Optimized detection pipelines to handle environmental noise, reducing false positive rates by 40% . – Improved system robustness against varied lighting through advanced preprocessing and feature extraction techniques.	Summer 2024

Achievements & Leadership

- Solved **500+ DSA problems** across LeetCode, GeeksforGeeks, and CodeStudio.
- **LeetCode Knight 1850 Max Rating**.
- **Infosys HackWithInfy 2026 Round 2**.
- **GATE 2026 Qualified · JEE Advanced 2022**.

Technical Member <i>Celestia – Annual Technical Fest of IITM Gwalior</i> – Created branding assets, posters, and digital creatives for technical fest outreach.	IITM Gwalior
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